UPPSALA UNIVERSITET

Department of Medical Sciences
SNP&SEQ Technology Platform

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REPORT ON GENOTYPING RESULTS

The SNP&SEQ Technology Platform at Uppsala University

REPORT issued by an Accredited Laboratory

Project name:	Internal_FU_KI
Project code:	XK-4162
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Date of delivery:	250520
Report issued by:	Project coordinator Susanne Björnerfeldt, Uppsala University, Dept. of Medical Sciences, Uppsala Biomedical Centre (BMC), D11:2, Box 1432, 751 44 Uppsala, Sweden Email: snp@medsci.uu.se
Analysis performed at:	The SNP&SEQ Technology Platform in Uppsala, Uppsala University, Dept. of Medical Sciences, Uppsala Biomedical Centre (BMC), D11:2, Box 1432, 751 44 Uppsala, Sweden
Accreditation:	Genotyping using the specified BeadChip type and chemistry version is an accredited method.

Result report contents:

Folder: XK-4162_250519_ResultReport
Description of result report:
XK-4162_250519_ResultReport.pdf

Manifest file:
GSA-24v3-0_A2.csv

Cluster file:
GSA-24v3-0_A1_ClusterFile.egt

Folder: XK-4162_250519_ResultReport_ICF_PLUS

Statistics from the genotyping:
XK-4162_250519_GenotypingStatistics_ICF_PLUS.xlsx

Genotype data:
XK-4162_250519_SNPGenotypeExport_ICF_PLUS.txt

Results from HapMap comparison:
XK-4162_250519_HapMap_1KG_Comparision_ICF_PLUS.xlsx

Identity test:
XK-4162_250519_Identify_Item.xlsx

Gender test:
XK-4162_250519_GenderEstimation.xlsx

PLINK files
Folder: XK-4162_250519_PLINK_ICF_PLUS
XK-4162_250519_PLINK_ICF_PLUS.map
XK-4162_250519_PLINK_ICF_PLUS.ped

IDAT files
Folder: XK-4162_250519_IDAT
XK-4162_250519_Samplesheet.csv
NOTE! The file format generated in the IDAT files is not covered by our quality accreditation ISO/IEC 17025.

Description:

Material&Methods
Number of analyzed SNP markers: 650 181
Number of analyzed samples or individuals: 22
Organism: *Homo sapiens*

The genotyping was performed using the Illumina Infinium assay and the results were analyzed using the software GenomeStudio 2.0.3.

BeadChip type:	GSA-24v3-0_A2
Manifest file:	GSA-24v3-0_A2.bpm
Genome build version:	38
Cluster file ICF:	GSA-24v3-0_A1_ClusterFile.egt

For reference to the Illumina Infinium assay see,
Gunderson KL, Steemers FJ, Lee G, Mendoza LG, Chee MS (2005) A genome-wide scalable SNP genotyping assay using microarray technology. *Nat Genet* 37:549-554

Steemers FJ, Chang W, Lee G, Barker DL, Shen R, Gunderson KL (2006) Whole-genome genotyping with the single-base extension assay. *Nat Methods* 3:31-33

Cluster files specify the borders for each created SNP marker genotype. Result files having the extension name **ICF** refer to genotyping based on cluster files generated by Illumina using a reference DNA sample set. Result files having the **PCF** extension refer to genotyping based on cluster files derived from the signal intensities from the DNA samples included in the project, generated at the SNP&SEQ platform. Illumina´s recommendation is to create and use reference cluster files based on signal intensities from the samples included in the project, (i.e PCF) when the number of samples exceeds approximately 100. When the number of samples in the project is fewer than 100, Illumina recommend using cluster files generated from at least 100 samples from normal subjects in another project, analyzed in the same laboratory and using the same chip type. If no PCF can be obtained Illumina recommend using ICF.

The result folders and files have the extension name ¨**PLUS**¨. This refers to that the file is based on genotype data exported from **PLUS** DNA strand according to the Human Reference Genome build38, as given in the marker file GSA-24v3-0_A2.csv.

Results

SNP conversion rate/ Fraction of SNP markers with sample call rate > 0:
99.36% (646 035 / 650 181)

4146 markers have sample call rate 0%
17.67% of all markers (114 879 / 650 181) have sample call rate >98%

Average sample call rate per SNP with sample call rate > 0:
88.35%

Fraction of non-polymorphic SNPs with sample call rate > 0
71.42% (461 411 / 646 035)

Reproducibility
99.99% (5 conflicts in 644 574 duplicate tests).
Number of markers with conflicts: 5

Fraction of SNP markers without duplicate error: 99.99% (646 030 / 646 035)

The value of reproducibility is calculated based on duplicate samples with SNP call rate ≥ 98% of the total number of SNP markers with sample call rate > 0%.
SNP markers with duplicate error(s) are listed under heading ¨Reproducibility¨ in sheet 2 and are marked in yellow in sheet 4, see column J, Duplicate failures in: XK-4162_250519_GenotypingStatistics_ICF_PLUS.xlsx

Average fraction of duplicate genotypings per marker
4.54%

Inheritance test in CEPH control samples
Not determined, no family information was available for the control samples.

Inheritance test in samples from the cohort
Not determined, no family information was available for the cohort.

HWE test
Hardy-Weinberg Equilibrium Chi2 test has been performed. SNP markers deviating from HWE are highlighted in yellow, see column I, HW chi2, in sheet 4 in: XK-4162_250519_GenotypingStatistics_ICF_PLUS.xlsx

Comparison of genotype data from Hap Map and 1000 Genomes vs genotype data generated in this project in CEPH QC samples
99.79% concordance (781 865 identical genotypes in 783 536 compared genotypes)

1 659 SNP markers with genotype data conflicts in one to two CEPH subjects in this comparison are listed separated in file
XK-4162_250519_HapMap_1KG_Comparison_ICF_PLUS.xlsx

One CEPH subject (Subject C16 [two replicates] in CEPH/Utah Pedigree 1362) included in this project were also included to generate the HapMap data. In addition, one CEPH subject (Subject C16 [two replicates]) from the same pedigree was included to generate the 1000 Genomes (1KG) data. On average 391 768 HapMap and 1KG SNPs per CEPH subject included for genotyping in this project were included in the comparison.

Note: HapMap and 1KG data should not be considered as key genotypes and for the markers identified with genotype conflicts we cannot say if the genotype from HapMap or the genotypes generated in this project are correct.

Average SNP call rate per sample:
87.79% (72.93-98.07%)

Note! Genotype data from samples with SNP call rate <90% have a high risk of being inaccurate and therefore should be considered to be excluded or used with great caution in further analysis.

Information on the SNP call rate per sample is given in sheet 3 in:
XK-4162_250519_GenotypingStatistics_ICF_PLUS.xlsx

Identity test
As part of our QC, the genotype data from each sample was pair wise compared to all other genotype data in samples included in the project. In this test we have used 200 markers with an allele frequency of approximately 0.5. 45 sample pairs above our threshold of 80% similarity were identified in this test and are listed in the file, XK-4162_250519_Identify_Item.xlsx

21 samples have low SNP call rate (<98%) which is known to impair the accuracy of this test.

Gender test
In the gender test, deviations between the given gender of the samples and the gender estimated from the genotype data for the samples are identified. In case no gender is given for a sample only the estimated gender will be shown. In 19 of the project samples the gender could not be determined based on the genotype data and the algorithm used by the software in GenomeStudio. In 18 of these samples the SNP call rate was lowered, <98%. A lowered SNP call rate is known to prevent the gender to be determined accurately. For results see:
XK-4162_250519_GenderEstimation.xlsx

Miscellaneous
Genotype data and QC statistics are exported from PLUS strand according to the Human Reference Genome.

Any coordinates for the SNPs given in this report as well as information reading PLUS DNA strand are based on the manifest file provided by Illumina, see GSA-24v3-0_A2.csv.

The results in this report apply to the condition of the DNA samples as received by the SNP&SEQ Technology Platform from the client and apply only to the DNA samples included in this report.

Samples with lowered SNP call rate (<98%) as well as SNP markers with lowered sample call rate (<98%) and/or that include duplicate errors and/or inheritance errors and/or conflicts with the HapMap data and/or have deviation from Hardy-Weinberg Equilibrium are more likely to include genotyping errors. The SNP&SEQ Technology Platform disclaims responsibility for the results of these samples and SNP markers.

Acknowledgement
In publications based on this data, the authors must acknowledge SciLifeLab and NGI with the following sentences:
"Genotyping was performed by the SNP&SEQ Technology Platform in Uppsala. The facility is part of the National Genomics Infrastructure (NGI) Sweden and Science for Life Laboratory. The SNP&SEQ Technology Platform is also supported by the Swedish Research Council and the Knut and Alice Wallenberg Foundation."

END OF REPORT